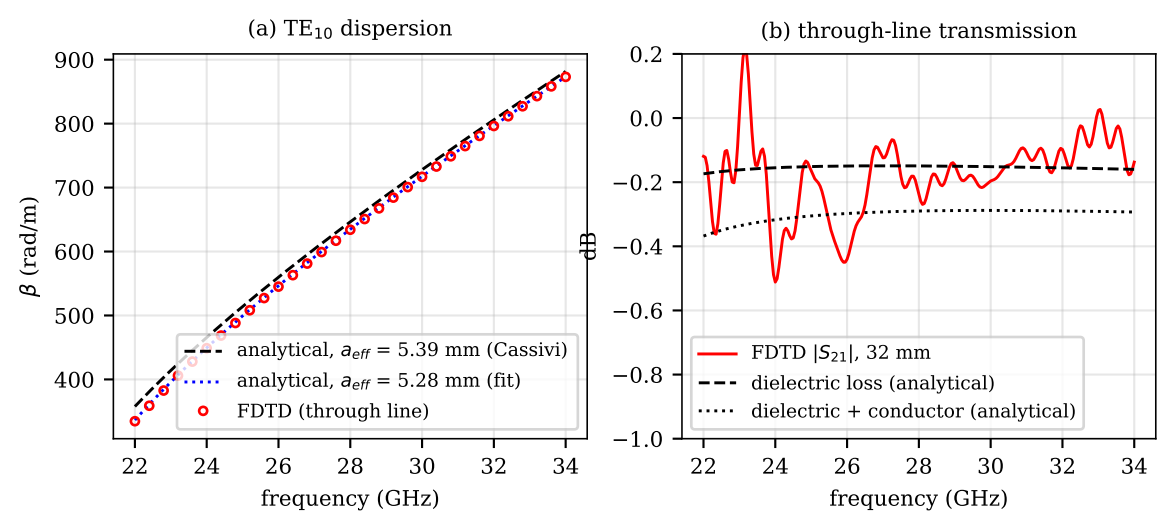
(a) TE10 dispersion
β (rad/m)
900
800
700
600
500
400
22
24
26
28
30
32
34
frequency (GHz)
analytical, aeff = 5.39 mm (Cassivi)
analytical, aeff = 5.28 mm (fit)
FDTD (through line)
(b) through-line transmission
dB
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
22
24
26
28
30
32
34
frequency (GHz)
FDTD |S21|, 32 mm
dielectric loss (analytical)
dielectric + conductor (analytical)